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## 1. An Introduction to Matrices

So, datastructures. By now, hopefully we're familiar with scalars and vectors. Scalars themselves have no dimension, and they can describe 1d values. The number line. Vectors themselves are one dimensional, and they can describe 2d+ values. Graphs.

But vectors are a datastructure which contains another datastructure. Its a storage type which stores scalars. What would happen if we had a vector containing other vectors, though? (Strictly speaking, we can't do this with vectors. Imagine for a moment, though, that we could.)

$$\begin{aligned}\vec{v} &= a\hat{i} + b\hat{j} = [a, b] \\ \vec{u} &= [c, d] \\ \vec{w} = [\vec{v}, \vec{u}] &= [[a, b], [c, d]] \approx \begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ or } \begin{bmatrix} a & c \\ b & d \end{bmatrix}\end{aligned}$$

Essentially, what we would have is a vector which is itself two-dimensional, regardless of the dimensionality of its data. Rather than a 2d vector, we have something like a  $2 \times 2$ d vector.

Well, we don't call it a vector. Vectors contain scalars; not other vectors. Instead, we call it a matrix.

A (term) matrix is just another type of datastructure, to borrow terminology from computer science. We can represent information using scalars, we can represent it using vectors, and we can represent it using matrices. Each of those tools have many associated applications and techniques.

(To extend the analogy further, we could also have matrices of matrices and so forth. But that's beyond the scope of this document here.)

—

(TODO) important to think of as separate from vectors too

Now, just as we can represent a scalar using a 1d vector, we can represent vectors as  $1 \times n$  or  $n \times 1$  matrices.

$$\vec{v} = a\hat{i} + b\hat{j} + c\hat{k} = \begin{bmatrix} a \\ b \\ c \end{bmatrix} = [a \ b \ c]$$

(TODO) where am I going with this

## 2. Binary Operations

Scalar-matrix multiplication and matrix addition/subtraction are trivial.

$$k \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$$
$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} a+e & b+f \\ c+g & d+h \end{bmatrix}$$

Notice that matrix addition is only defined between equally-sized matrices.

Matrix-scalar addition is formally undefined, although it is sometimes useful to allow:

$$k + \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} k+a & k+b \\ k+c & k+d \end{bmatrix}$$

Matrix multiplication is more complicated and explained below.

Matrix division is treated as an extension of matrix multiplication, in the same way that scalar division is multiplication by a negative exponent. It will be discussed separately as the “Inverse”.

### 2.1. Matrix Multiplication

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \times \begin{bmatrix} e & f \\ g & h \end{bmatrix}$$
$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} \neq \begin{bmatrix} ae & bf \\ cg & dh \end{bmatrix}$$
$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{bmatrix}$$
$$\begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = [ac+bd]$$

(technique) The procedure to do matrix multiplication is as follows:

0. Given  $AB = C$ , find  $C$ .
1. Know that  $C$  will have the height of  $A$  and the width of  $B$ .
2. The  $(i \times j)$ th element of  $C$  will be the summation of the piecewise multiplication of the  $i$ th row of  $A$  with the  $j$ th column of  $B$ .
3. Repeat the prior step for each element in  $C$ .

From this we can see that matrix multiplication is defined if and only if the width of  $A$  equals the width of  $B$ .

Further, recognize that matrix multiplication is not commutative.

$$AB \neq BA$$

#### 2.1.1. The Identity Matrix

$$[1], \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \dots$$

The **(term) identity matrix** is a matrix with 1s along its diagonal and the rest zero, as shown above. It is usually denoted with  $I$ , or sometimes  $I_n$  where it is an  $n \times n$  matrix.

Consider that, with scalar-scalar multiplication, we have the concept of “1”, where anything times 1 is equal to itself. The identity matrix is matrix-multiplication’s equivalent. (You can do the matrix multiplication to show this, if you’d like.)

$$AI = A$$

$$IA = A$$

### 3. Unary Operations

#### 3.1. The Determinant

let  $A$  be any square matrix

$$\text{determinant of } A = \det(A) = |A| = \begin{vmatrix} a & b & \dots \\ c & d & \\ \dots & & \end{vmatrix}$$

The **(term) determinant** of a matrix is like its “magnitude”, borrowing the concept from vectors. It’s a little harder to visualize, though, since matrices are multidimensional compared to a vector’s 1d shape. Consider, though:

$$|CA| = C|A|$$

$$|AB| = |A| |B|$$

Note, also, that only square matrices have determinants.

Unfortunately, computing determinants is harder than vector-magnitudes. We define the process recursively. First, the “base cases” are the trivial determinants of  $1 \times 1$  and  $2 \times 2$  matrices:

$$|[a]| = a$$

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - cb$$

To remember the  $2 \times 2$  matrix, consider that we just multiplied across the diagonals and subtracted the terms.

I will present the  $3 \times 3$  and  $4 \times 4$  determinants before explaining the process below.

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

$$\begin{vmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{vmatrix} = a \begin{vmatrix} f & g & h \\ j & k & l \\ n & o & p \end{vmatrix} - b \begin{vmatrix} e & g & h \\ i & k & l \\ m & o & p \end{vmatrix} + c \begin{vmatrix} e & f & h \\ i & j & l \\ m & n & p \end{vmatrix} - d \begin{vmatrix} e & f & g \\ i & j & k \\ m & n & o \end{vmatrix}$$

**(technique)** Now, to compute the determinant:

0. We are given some  $n \times n$  matrix and asked to find its determinant, where  $n \geq 3$ .
1. Pick some row or column as our top-level scalar multipliers. (In the examples above, the first row (abcd) was chosen both times. We could have just as easily picked efgh or cgko.) We will iterate through that row/column.

- Each item in that row/column corresponds to a term in the output. The term consists of that item multiplied by the determinant of the  $(n - 1) \times (n - 1)$  matrix formed by omitting that item's row and column.
- Select the correct sign for the term according to the checkerboard  $\pm$  pattern explained below.
- Repeat for each item in the initially selected row/column.
- We now have a problem involving  $(n - 1) \times (n - 1)$  matrices. Repeat until we have  $2 \times 2$  matrices which we can resolve with the known formula.

**(TODO)** better visuals

The checkerboard pattern is demonstrated:

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix} \quad \begin{bmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{bmatrix} \quad \begin{bmatrix} + & - & + & - & + \\ - & + & - & + & - \\ + & - & + & - & + \\ - & + & - & + & - \\ + & - & + & - & + \end{bmatrix}$$

**(technique)** Consider that, since we can select any row/column as our starting point, it is advantageous to select that row/column which contains the most zeroes, since we know a zero's corresponding term will zero-out.

### 3.1.1. Implications of the Determinant **(TODO)**

## 3.2. The Transpose **(TODO)**

let  $A$  be any matrix  
 transpose of  $A = A^T$   
 if  $B = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}$  then  $B^T = \begin{bmatrix} a & c & e \\ b & d & f \end{bmatrix}$   
 $(A^T)^T = A$

**(TODO)** symmetric, skew symmetric

## 3.3. The Adjugate (or Classical Adjoint) **(TODO)**

let  $A$  be any square matrix  
 adjugate of  $A = \text{adj}(A)$

The terms **(term)** adjugate and **(term)** classical adjoint are synonyms.

## 3.4. The Trace **(TODO)**

## 3.5. The Inverse **(TODO)**

## 4. Row Operations **(TODO)**

## 5. Eigenvalues and Eigenvectors **(TODO)**

## 6. Laplace Transformations **(TODO)**